Gupta *et al. Journal of Big Data* (2025) 12:256
https://doi.org/10.1186/s40537-025-01300-9

Journal of Big Data

RESEARCH Open Access

A multimodal fusion model for real-time environment emotion recognition using audio-visual-textual features

Chhaya Gupta[1,2], Nasib Singh Gill[1*], Preeti Gulia[1*], Abhinav Kumar[3,4*], Hanen Karamti[5], Demmelash Mollalign Moges[6*] and Imen Safra[7]

*Correspondence:
Nasib Singh Gill
nasib.gill@mdurohtak.ac.in
Preeti Gulia
preeti@mdurohtak.ac.in
Abhinav Kumar
drabhinav.urfu@gmail.com
Demmelash Mollalign Moges
demmelashm@outlook.com

Full list of author information is available at the end of the article

Abstract
Multimodal combines multiple modalities to create insightful conclusions or to make more precise predictions. Nowadays, the multimodal concept is used to identify human emotions precisely. This study proposes a three-stage novel M-fusHER (Multimodal fusion Human Emotion Recognition) multimodal model for human emotion recognition in real-time with the help of text, audio, and videos. In the first stage, features are extracted with the help of a convolutional neural network merged with multiplicative LSTM. In the second stage, video and audio data, text, and audio are fused in binary form. In the third stage, real-time object detection for human emotion recognition on real videos is implemented. The experimental results are obtained by fusing audio, text, and videos by considering the standard features. For object detection, a fine-tuned YOLOv6 model was used for detecting facial features and expressions from the video. The multiplicative LSTM is also used to extract and learn from the text features. Three datasets, i.e., IEMOCAP, MOSEI, and MELD are used for implementation, and the detection accuracy of the proposed model M-*fus*HER on IEMOCAP, MOSEI, and MELD datasets is 95.45%, 88.76%, and 95.41% approximately, which is quite encouraging.

Keywords Audio processing, Convolutional neural network (CNN), Multiplicative LSTM, Multimodal modeling, Text processing, Video processing

Introduction

One of the cornerstones of human-computer emotional connection is emotion recognition. The difficulty in recognizing emotions is the restricted amount of emotional information in a single mode. One-dimensional information is not remarkably realistic compared to real emotions because people spontaneously communicate and express their feelings differently. Due to noise and the absence of additional dimensions of emotion detection, one-dimensional information only analyses one degree of emotion, which may not accurately represent true feelings. Another issue is the availability of tagged data, which might be challenging to obtain as it is either noisy sometimes or unavailable.

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Hence, a multimodal approach to analyzing and recognizing emotion is the main focus of current research.

Emotion recognition is an essential component of research on human-computer emotional connection and also impacts the advancement of AI technology [1]. The advancement of emotion recognition research has accelerated the development of other fields. Emotion recognition research focuses on text, audio, and image monomodal emotion recognition models. However, unimodal emotion recognition has improved in recent years, but with some limitations. In its present state, the unimodal emotion recognition network does not convey a user's feeling properly, but combining it with other modal elements will result in a more in-depth description of the emotion [2]. Hence, multimodal modeling has become a critical research model for human emotion recognition [3].

Online videos are becoming the most common sort of data that people use to communicate and share activities in the digital realm. Video data analysis has undoubtedly received much attention from the scientific and business communities, particularly in video sentiment assessment and emotion recognition. People use a variety of strategies to express their feelings. The main ways that people express their emotions are through language, voice, and facial expression. Furthermore, how people blend these methods to communicate their feelings is intricate. In some situations, predicting human emotions solely from one modality data might result in inaccurate predictions. To make the phenomenon clear, let us take a few examples. Suppose a wildlife photographer traveling through Air India lost his luggage with his camera. He went to the airport staff and filed a complaint for misplacement of his things. But after arguing with them for a long time, he understood that the airport staff wasted his time and was informed that his luggage had been stolen. He angrily left, saying, "Thank you very much" [4]. Now, how can the emotion behind this sentence be measured? The anger behind this sentence can be measured by the person's facial expressions and tone of voice. If the sentence is read-only, the reader is confused about why the person is grateful to the airport staff. Hence, emotional recognition becomes essential to understand the meaning behind this sentence. Hence, creating an efficient fusion technique for several video data to produce a reliable joint representation is still problematic research.

This paper presents a fusion method based on the architecture of the transformers in multimodal deep learning. This method effectively fuses multimodal features and learns an integrated representation from the unprocessed video data for multi-label emotion recognition. In particular, the suggested model accepts inputs from text transcripts from the video, audio waveforms, and unprocessed video frames. The model employs each modality's appropriate deep neural architectures to extract high-level features from the unprocessed input. Then, it uses the transformers' multi-head attention mechanism to develop a robust joint multimodal representation of the multimodal features. The transformer's multi-head attention mechanism inspects every component of the input sequence, learning a more precise feature sequence that highlights significant components and fades out unnecessary ones. The process of gathering data, preprocessing the data, extracting emotional features from the data, and categorization are all included in emotion recognition. The progressive development and application of multi-modal fusion in recognizing emotions aim to fully utilize the available data while

capturing pertinent information across and within modalities. This paper proposed a novel M-fusHER model. The main contributions of this study are:

- To attain high detection accuracy and utilize every single-modal feature for emotion recognition, a novel three-stage M-*fus*HER model is proposed. The model combines a multi-head attention module in the transformer encoder that helps perform cross-attention and integrate information at the token level.
- The M-*fus*HER model has three stages: feature extraction, data fusion, and real-time emotion recognition. In the first stage, features are extracted with the help of a convolutional neural network that is merged with multiplicative LSTM. In the second stage, audio, video, and text data are fused in binary format. In the final stage, real-time emotion recognition takes place with the help of a finetuned YOLOv6 model.
- All the stages are combined to form a single neural network. The results indicate that the suggested model outperformed the others in terms of accuracy compared to existing studies.

The subsequent sections are structured as follows: A review of relevant literature on multimodal emotion recognition is provided in the second section. The suggested model M-fusHER is presented in the third section. The fourth section is about implementation and outcomes, and it includes an ablation study and an evaluation of the suggested model's performance. Section five finally concludes the paper.

Related literature survey

Multiple media are needed to present and adequately identify emotions in human-computer interaction. The durability of emotion recognition can be enhanced because several modalities cover more information than single modalities. Multimodal emotion recognition technology considers how text, speech, images, and other modalities interact. This section presents a brief literature survey of all the work done till now in the stated field in Table 1.

More recent developments in multimodal learning have brought many new architectures into the field of emotion recognition. One of them is the RMER-DT, in which the diffusion and transformer-based model of robust multimodal emotion recognition in dialogues is presented, and generative diffusion models are proposed as the solution to optimize cross-modal correspondence [23]. And like RAFT (Robust Adversarial Fusion Transformer), adversarial fusion has been shown to effectively enhance sentiment analysis, which is also comparable to our objective of enhancing robustness under noisy or missing modalities [24]. CIME (Contextual Interaction-based Multimodal Emotion Analysis) and contrastive-based negative information removal is also a source that substantiates a primary contribution that context modeling and contrastive goals make to the semantic representations that the multimodal emotion analysis task comprises [25].

In addition to emotion recognition capabilities, the development of the nearby areas suggests the significance of robustness, generalization, and applicability in a real-time setting. Dynamic global local spatiotemporal graph air quality prediction, and knowledge generation with distillation road segmentation in intelligent transportation systems are examples of greater complexity in temporal relationships and changeover of domains [26]. Similarly, federated domain generalization ophthalmology approaches focus on

Table 1 Brief related work with research gaps and future scope in multimodal emotion recognition

Author	Dataset	Model	Results	Research gap
Mingyong Li et al. [5]	IEMOCAP dataset	Multiobjective multimodal emotion recognition model with DCNN for speech emotion recognition and DSCNN for video emotion recognition	The model achieved an accuracy of approximately 71%.	Real-time scenarios are not used to evaluate the model.
Hiranmayi Ranganathan et al. [6]	Custom dataset, Cohn Kanade dataset, mind reading database, DEAP database, MAHNOB-HCI dataset	The study presents four DBN models and proposes a convolutional DBN model for emotion recognition	The CDBN model achieved accuracies of 83%, 94%, 92%, 79%, and 58% on different datasets, respectively.	The model is still waiting to be tested on real-time scenarios.
Xia Liu et al. [7]	IEMOCAP and MOSI dataset	The CMC-HF (Cascaded Multi-channel and hierarchical Fusion) model is proposed	The proposed CMC-HF model achieved an accuracy of approximately 88%.	Feature extraction is still an open challenge in this study.
Ying Hu et al. [8]	DEAP dataset, Custom dataset	A spatio-temporal neural network is proposed for emotion recognition with the help of face and EEG signals.	The model attained an approximate accuracy of 84%.	The model is insufficient for real-time scenarios.
Yuanyuan Wang et al. [9]	IEMOCAP and MELD datasets	A novel approach called multi-modal transformer-augmented fusion is suggested for speech emotion recognition.	The model obtained an accuracy of 72%.	The model is insufficient in detecting facial expressions from videos.
Aayushi Chaudhari et al. [10]	RAVDESS dataset	Proposed an inter-modality attention (IMA) token-based model for recognizing emotions.	The model achieved an accuracy of approximately 87%.	The study claims to examine emotions in contextual data in the future.
Yujin Wu et al. [11]	PRESAGE, WESAD, CAGE, K-Emocon datasets	Profounded a self-supervised learning model for emotion recognition in wearable devices	The model attained an approximate accuracy of 84%.	The model is trained on limited data so far.
Jintao Wen et al. [12]	IEMOCAP and MELD dataset	Suggested a Dynamic Interactive Multiview Memory Network (DIMMN) model to recognize emotions with the help of interactive information	The model achieved an accuracy of approximately 64%.	The model can be fine-tuned further for better accuracy.
Aditi Sharma et al. [13]	IEMOCAP and MELD datasets	The Profounded EmoHD model is based on Ekman's model for emotion recognition.	In terms of accuracy, the model performed well.	The model is trained on the raw dataset and can be finetuned in the future to attain better results.
Yurui Xu et al. [14]	CH-SIMS dataset	Suggested a dual-modal fusion model for emotion recognition by integrating features of audio and text	The model achieved an accuracy of approximately 90%.	The model has not been tested on video outcomes and can be finetuned further to attain better results in the future.
Esam Ghaleb et al. [15]	RAVDESS, CREMA-D dataset	Profounded bi-modal time windows spanning a short video model, namely, MATER	The model achieved good results in terms of accuracy.	The model will be further enhanced with attention modules and custom datasets.
Yansen Wang et al. [16]	IEMOCAP dataset	Proposed a Recurrent attention model named RAVEN	The model achieved good results.	The model could not perform well when the sequences were extracted at a high frequency.

Table 1 (continued)

Author	Dataset	Model	Results	Research gap
Nicolas Audebert et al. [17]	RVL-CDIP, Tobacco3482 datasets	Worked with MLP	The model achieved good results working only with text and image datasets.	The model cannot identify the missing modalities and is not trained in the modalities that are influenced for any given sample.
PanagiotisTzirakis et al. [18]	SEWA dataset	Profounded a model that is a combination of self-attention, hierarchical attention, and LSTM	The model achieved good results.	Traditional LSTM is not a suited classifier for non-sequential features; hence, the model cannot correctly identify the modalities.
Dushyant Singh Chauhan et al. [19]	CMU-MOSEI dataset	Suggested a Bi-GRU neural network	In terms of accuracy, the model performed well.	The model has not undergone training for real-time environments.
Juanpablo Heredia et al. [20]	IEMOCAP dataset	Profounded the model based on probability, namely, Embracenet+	In terms of accuracy, the model performed well.	In this model, the overall model's performance depends on each modality's performance.
Dou Hu et al. [21]	IEMOCAP, MELD datasets	A new multimodal dynamic fusion network (MM-DFN) is suggested for recognizing emotions in conversations.	The model achieved good results.	The model is not trained for a real-time environment.
Feiyu Chen et al. [22]	ERC dataset	A novel Graph Neural Network (GNN) based model is introduced to recognize conversations' emotions.	The model achieved good results.	The model cannot identify the missing modalities and is not trained in the modalities influenced by any given sample.

privacy-sensitive and general training systems, aligning with the future of multimodal emotion recognition on distributed systems as well [27]. The recent focus on reliable deployable AI can be seen in recent developments in credible image super-resolution and client/server-based recognition systems. Lastly, spatial-kernel-selection-based deepfake detection, halo-attention-based deepfake detection, and feature-refinements-based deepfake detection are more of an attention-seeking model that aims to make use of attention-based architecture in solving multimodal perception tasks.

The literature survey depicts the limitations of the SOTA methods used so far. Multimodal visual, textual, and audio fusion for video data is the main emphasis of this work. Instead of dividing several modalities into pairs or a trio combination, a single transformer-based architecture integrates the data from all modalities simultaneously.

Proposed methodology

The goal of multi-label emotion recognition with a given input video segment 'V' having several modalities, such as video frames, text transcription, an audio signal, and a list of E emotions that has been predetermined, is to forecast whether each emotion will appear in the video or not. The overall structure of the profounded model is depicted in Fig. 1. The suggested model's architecture illustrates the multilabel multimodal emotion identification method, which accepts a video clip as input and classifies the emotions displayed in the clip. Figure 1 is explained in the further section of this study.

The proposed model is composed of three stages: feature extraction stage, fusion stage, and embedding stage. CNN-mLSTM (Convolutional Neural Network combined with multiplicative LSTM) extracts features for encoding text translations, video frames,

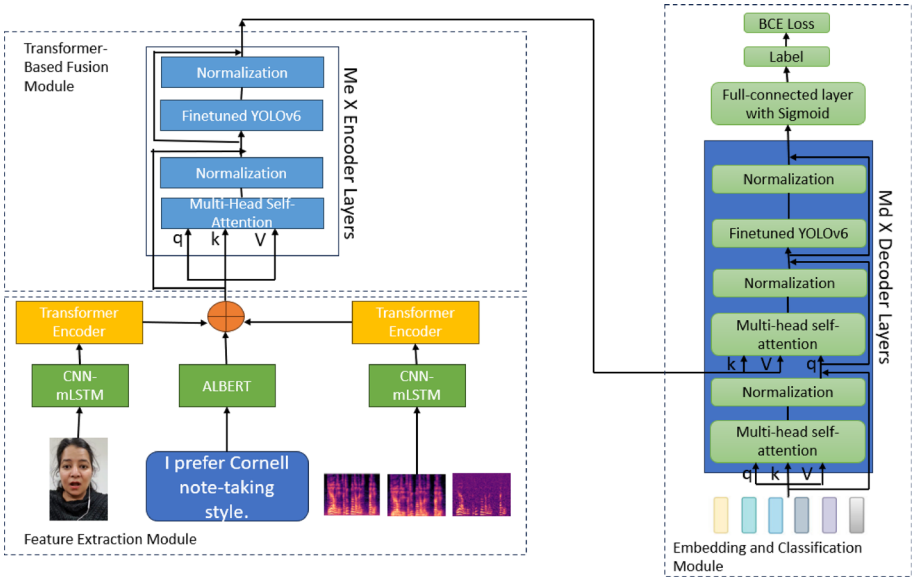

Fig. 1 The architecture of the proposed model M-*fus*HER. The three modules—the Transformer-Based Fusion, the Embedding, and the Feature Extraction modules (FEM)—are shown in the architecture. Text, audio, and visual modalities' features are extracted via the FEM. The transformer-based fusion module (TFM) fused the multimodal data. The embedding module does emotion identification and classification

and audio signal inputs [28]. In the fusion stage, different modalities are fused with the help of a Multi-Head Attention Transformer Encoder (MHATE). In the embedding stage, the transformer decoder extracts the emotions from the fused modalities. Finally, a finetuned YOLOv6 model is executed to train the model as per the extracted features and make predictions of emotions. All the stages are discussed in detail further in this section.

Raw audio was resampled to 16 kHz and put into zero mean/unit variance. Short-Time Fourier Transform (STFT) with a window length of 25 ms and a 10 ms hop (i.e., 400 samples window, 160 samples hop at 16 kHz) was used. We scale the resulting magnitude spectrogram to a Mel scale with n mels = 64, and use a log amplitude scale: log(1 + 10 * mel) (or log(mel + eps) - indicate which in code). As model input, we divide audio into 400 ms analysis blocks (non-overlapping to allow indexing), and, in each block, we stack the sequence of Mel frames generated by the 10 ms hop. This produces a chunk of ~ 40 Mel frames per 400 ms, which is seen as a single 2-D image (time × mel), and fed into the CNN-mLSTM encoder. The feature extraction was done by sampling the videos at 1 sample/s. On every frame that we sample, we run the fine-tuned YOLOv6 face detector. YOLOv6 detections with confidence 0.5 or above are kept, the visual bounding box of the detected face is cropped and resized to 64 × 64 (RGB) and normalized to the range [0,1], and run through the CNN-mLSTM visual backbone. When more than one face is found, we retain the largest/closest face (maximal area) to the desired target subject, and when no face is found at all, we assign the visual availability flag to 0 at that time-step (modal dropout). One more tradeoff with the RTX8000 was the 64 × 64 crop, since only the RTX8000 could perform inference in real time, but with that small crop.

At chunk granularity, temporal alignment is carried out both between audio and video. Individual audio pieces of 400 ms are synchronized with the video stages sampled over that very 400 ms. Since video runs at 1 Hz (1 frame/sec), we upsample video characteristics to audio chunk rate by storing the latest frame of video as the exact visual encoding

over the 400 ms interval (frame-replication). Instead (more recommended to align them better), record a sample video at 25–30 fps and choose the frame(s) the timestamps of which coincide with the audio chunk - we suggest the latter option in case you use the code. The text of the paper already explained the 10 ms Mel hop, the chunking 400 ms; the correlation above makes clear that the 10 ms hop generates the internal Mel frames, whereas the 400 ms is the chunk size at a higher level of fusion.

Feature extraction

This study aims to enhance learning representations and develop a multimodal fusion for video data to improve model performance. In this stage, CNN merged with multiplicative LSTM (CNN-mLSTM) is used for feature extraction [28]. The CNN model performs Convolutions on input data using three filters: 1 × 1, 3 × 3, and 5 × 5. The output of the classification layer is fed into multiplicative LSTM layers. A multiplicative recursive neural network (mRNN) and an LSTM are combined to form a multiplicative LSTM [29]. The three gates of an LSTM are the Input gate I, the forget gate F, and the Output gate O. The Previous hidden state, ht-1, and the input layer, yt, are given as input to the current hidden state, ht, as depicted in Eq. 1.

$$\widehat{h_t} = w_{hy} y_t + w_{hh} h_{t-1} \tag{1}$$

where, $\widehat{h_t} \in \mathbb{R}^k$ is the current state at time t, $y_t \in \mathbb{R}^d$ is the input vector at time t, $h_{t-1} \in \mathbb{R}^k$ is hidden state from previous time t-1 $w_{hy} y_t$ is the hidden state's weight w.r.t input layer y, $w_{hh} h_{t-1}$ is the weight of the previous hidden state.

The three gates are stated as:

$$I_t = \partial \left(w_{Iy} y_t + w_{Ih} h_{t-1} \right) \tag{2}$$

$$O_t = \partial \left(w_{Oy} y_t + w_{Oh} h_{t-1} \right) \tag{3}$$

$$F_t = \partial \left(w_{Fy} y_t + w_{Fh} h_{t-1} \right) \tag{4}$$

where, ∂ = sigmoid function, w = weight.

The relationship between the input and output gate's component sections determines what data should be stored and deleted at each transition. The input gate creates an internal state vector s_t to specify how much input should be sent to each hidden unit. This internal state vector is XNOR with a forget gate, and the previous internal state vector is XNOR with an output gate and current hidden state. Forget gate F aids in calculating the amount of past internal state preserved as s_{t-1}. The internal state is stated as follows;

$$s_t = F_t \odot s_{t-1} + O_t \odot tanh\left(\widehat{h_t}\right) \tag{5}$$

Information that may not be needed right now but will be required later on is preserved with the help of the output gate. In multiplicative LSTM, an intermediate state b_t is combined with each gate, as shown in the following equations. The intermediate state is used in a mLSTM to introduce a multiplicative gating mechanism, which helps model more complex interactions between the input and hidden states. This state in mLSTM captures more intricate relationships by combining both the hidden state and the input

multiplicatively. This state decides how much input and hidden state should contribute to the next hidden state.

$$b_t = (w_{by} y_t) \odot (w_{bh} h_{t-1}) \tag{6}$$

$$\widehat{h_t} = w_{hy} y_t + w_{hb} b_t \tag{7}$$

$$I_t = \partial \left(w_{Iy} y_t + w_{Ib} b_t \right) \tag{8}$$

$$O_t = \partial \left(w_{Oy} y_t + w_{Ob} b_t \right) \tag{9}$$

$$F_t = \partial \left(w_{Fy} y_t + w_{Fb} b_t \right) \tag{10}$$

Transformer-based encoder for multimodal fusion

Transformer-based multimodal fusion (TMMF) uses a transformer architecture to integrate information from multiple data modalities (e.g., text, image, audio, video, etc.). The goal is to leverage the strengths of transformers, such as their self-attention mechanism, to effectively combine different types of data and capture relationships between them. The transformer architecture excels at capturing relationships between elements in a sequence with its self-attention mechanism. In multimodal fusion, the idea is to extend this capability to capture interactions across different modalities. Transformers learn how different types of data contribute to a common understanding of the input, and they use cross-modal attention to identify relationships between, for example, an image and a text description. Self-attention allows the model to weigh the importance of different input elements (words, pixels, etc.), and cross-attention layers are used to align data from various modalities. For instance, a cross-attention mechanism might learn which words in a sentence refer to which objects in an image or video frame. This study proposes a novel features-fusing technique for the synchronous fusion of features of various modalities.

Given an input sequence as T € N^{v*s} having s size of v vectors, the multi-head self-attention layer (MHSAL) of the transformer encoder simultaneously multiplies T to a set of three components, namely, query, key, and value; $q_j \epsilon N^{v*s_k}$, $k_j \epsilon N^{v*s_k}$, and $V_j \epsilon N^{v*s_v}$ respectively, in m different subspaces where several heads are m and usually, s_k=s_v. A single attention function is implemented on all these sets, shown in Eq. 11.

$$Att\left(q_j, k_j, V_j\right) = softmax\left(\frac{q_j k_j^T}{\sqrt{s_k}}\right) V_j \tag{11}$$

$$softmax\left(y_i\right) = \frac{exp\left(y_i\right)}{\sum_j^i iexp\left(y_j\right)} \tag{12}$$

Where the product $q_j k_j^T$ is the similarity measure, and the $Att\left(q_j, k_j, V_j\right)$is the attention weight, which gives the weighted sum. This attention weight is also known as the SoftMax score as depicted in Eq. 12. The final representation is obtained by taking an average of all attention head outputs along with trainable weights as:

$$Multihead\left(q, k, V\right) = concat\left(head_1, \ldots, head_m\right) L^0,\ head_j = Att\left(q_j, k_j, V_j\right),\ j\epsilon 1, \ldots, m \tag{13}$$

where $L^0 \epsilon N^{ms_v*s}$ = learnable parameters.

The model can learn which elements it should first attend by scanning through each one in the input sequence with a single dot product attention. With the help of multiple head attention modules, the process can be executed from various subspaces. In other words, the transformers offer a way to gather data about the output while selectively accumulating knowledge about the complete input sequence. Moreover, the vector order in the input sequence does not affect the multi-head attention. Since the input sequence comprises an order-agnostic mixture of data from different modalities, it makes sense to employ multi-head attention to fuse multimodal information.

From the feature extraction stage, three sequences have been obtained, namely, text translations as $T\epsilon N^{v_t*s}$, image frames from the video as $I\epsilon N^{v_i*s}$, and audio signals as $A\epsilon N^{v_a*s}$. All these are concatenated together to form a multimodal feature sequence. These multimodal features are initially added with a classification token (CLST). Traditional transformer encoders consist of components such as MHSAL, normalization layers (NL), and a feed-forward network (FF). After this, these sequences are used as input for the fusion module. The fused multimodal (FM) is represented as $FM^i\epsilon N^{v_j*s}$ where, $v_j = v_t + v_i + v_a$. The FM^iat a block, *i is* calculated as;

$$FM^i_{temp} = NL\left(MHSAL\left(FM^{i-1}\right) + FM^{i-1}\right) \tag{14}$$

$$FM^i = NL(FF\left(FM^i_{temp}\right) + FM^i_{temp}) \tag{15}$$

where, $i\,\epsilon\,1, \ldots, M_e$ and $F^0 = concat(E_{CLST},\, T, I, A)$.

In this study, a tuned vanilla transformer encoder is used that is stacked at M_e multiple blocks to create the multimodal fusion module [30]. The classic vanilla encoder was unable to enhance the overall architecture's performance. Hence, parameters like embedding vector dimensions, number of heads, hidden layer size of FF in transformer, vocab size, maximum sequence length, and dropout probability are tuned to improve the suggested model's overall performance. The abovementioned parameters can be increased or decreased as per the dataset used.

In this study, the fusion algorithm used is shown in Algorithm 1. The proposed model M-*fus*HER receives all the data from all modalities, such as features from face videos, text translations, and audio data in a Mel spectrogram. A Mel spectrogram for multimodal refers to using Mel spectrograms as a representation of audio data in multimodal systems that integrate data from multiple modalities such as text, images, audio, video, etc. In multimodal systems, different data types (like images, video, and text) provide complementary information. In a video, the visual frames (image modality) are combined with the Mel spectrogram (audio modality) to understand both what's happening in the scene visually and what's being said or heard. The Mel spectrogram represents the sound in speech-to-text or audio-driven text generation systems, while the text modality provides the linguistic context. In multimodal fusion models (e.g., using a transformer-based multimodal fusion system), Mel spectrograms are used alongside other features like visual or text embeddings. For instance, in a video, the model can learn how certain sounds correspond to particular objects or emotions in the visual frames. This study uses Mel spectrograms alongside video frames of facial expressions to predict emotions. The audio provides tone, pitch, and intensity cues, while the video offers visual facial expressions, allowing the model to make more nuanced predictions. The Mel spectrogram is

extracted from raw audio using standard signal processing techniques (Short-Time Fourier Transform, Mel scale conversion, logarithmic scaling). It is often treated like a 2D image, with time on the x-axis, frequency on the y-axis, and color intensity representing amplitude. This makes it suitable for processing by Convolutional Neural Networks (CNNs) feature extractors used in multimodal pipelines. After generating the Mel spectrogram, it is passed through CNN-mLSTM to produce a lower-dimensional embedding, which can then be fused with embeddings from other modalities like text or image. The main challenge faced was aligning the Mel spectrogram with other modalities, like video frames, which requires careful synchronization to ensure that both modalities contribute relevant information simultaneously. Mel spectrograms have a different structure (2D, like images) compared to text (1D) or video (3D), so careful feature extraction and normalization are needed to handle these differences in multimodal systems. When synchronizing audio with video, the time axis of the Mel spectrogram (which corresponds to the duration of the audio) must align with the time axis of the video frames. Mel spectrogram is computed at regular intervals (e.g., every 10 ms) to do this. The video may be sampled at a similar frame rate (e.g., 30 frames per second), allowing audio and video data to be split into the same number of time steps. If the video and audio have different time resolutions, one modality can be down-sampled or up-sampled to match the other. After extracting features from each modality, the next step is to fuse them meaningfully so that the model can learn their relationships. In this approach, the model uses self-attention layers to selectively focus on relevant parts of one modality based on the information from another modality. All the models used in this study are first initialized, and an extra linear layer is added to get the output at the end. The vector known as the availability vector is instantiated. This vector helps in assessing each modality. The vector is initially assigned a value of 1. This value changes to 0 if there is no input data. All the probabilities that are required for this fusion are set.

Input: Facial video predictions, text translation, Mel spectrogram
Procedure:

```
All the individual models are initialized: CNN-mLSTM, finetuned YOLOv6
The extra linear layer is added.
Available_v← [1,1,1]
If face_pred == NULL then
Available_v[0] ← 0
Face_pred← Finetuned_YOLOv6(image_frames)
End if
If text_translation == NULL then
Available_v[1] ← 0
Text_pred← ALBERT(translation)
End if
If Mel_spec == NULL then
Available_v[2] ← 0
Audio_pred← Finetuned_YOLOv6(melspectogram)
End if
Probability_1 ←set_prob()
Modal ← [Avg(Face_pred, Text_pred, Audio_pred)]
Concat_modal← concatenation(Modal, Available_v)
Return M-fusHER(Concat_model)
```

Algorithm 1: Fusion process for M-fusHER

Emotion embedding

Unlike earlier transformer-based studies that typically utilize the classification token's outcome for straightforward categorization using flat layers, this approach utilizes the entire output sequence of the transformer encoders of the fusion module to improve the multi-label emotion identification features, as illustrated in Fig. 2. This methodology

embraces the idea of acquiring numerous embeddings, with each embedding customized for a specific emotion.

Several emotion-level embeddings are combined by using the transformer decoder's cross-attention module. In this stage, the output of the fusion stage $FM\epsilon N^{v_j*s}$ is taken as input, and it generates the emotion embeddings as $E \in N^{c*fd}$ (where c is the number of emotion classes and fd is the size of the feature dimension). In this stage, an emotion-specific embedding is first initialized, which is used to produce the query vector q_j. The key and value vectors are simultaneously projected using the enhanced multimodal characteristics sequence. The video emotions are learned with the help of transformer decoders and the multi-head self-attention layer (MHSAL) as:

$$E_j^{!} = NL(E_{j-1} + MHSAL\,(E_{j-1}, E_{j-1}, E_{j-1})) \tag{16}$$

$$E_j^{!!} = NL(E_j^{!} + MHSAL\left(E_j^{!}, FM, FM\right)) \tag{17}$$

$$E_j = NL(E_j^{!!} + FF\left(E_j^{!!}\right)) \tag{18}$$

YOLOv6 is used to classify objects in images and videos. This version of YOLO performs better than all the other versions available. In this study, a lightweight YOLOv6 network for object detection is used. The model is lightweight because a hidden layer pruning technique reduces the parameters and depth of the model. However, because of the pruning procedure, the model's detection accuracy degrades; as a result, a transfer learning strategy is employed to improve the network's detection accuracy [31]. YOLOv6 consists of a neck, a decoupled head, and a backbone. Varifocal loss and distribution loss functions are the two loss functions used in this version of the YOLO model [28]. Pre-trained YOLOv6 models are typically trained on large, generic datasets like COCO, which may not include the specific objects or features that must be detected. Fine-tuning allows the model to adapt to a specific custom dataset. It can improve accuracy for a particular environment (e.g., detecting objects in low-light conditions, underwater, or in specific weather). Depending on the complexity and size of the custom dataset, the layers of the model are adjusted (e.g., changing the head, anchor size, or even making

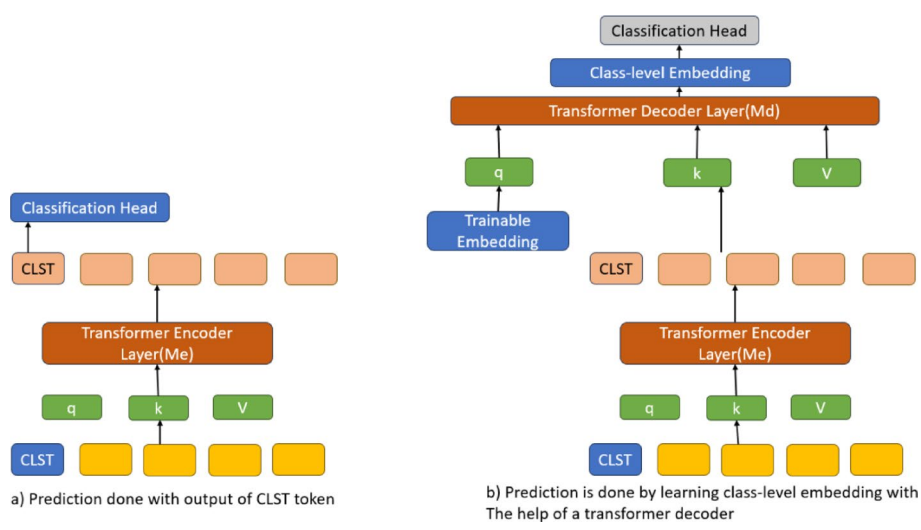

Fig. 2 Illustration of predictions made by the CLST token and with the help of a transformer decoder

the architecture deeper for small objects or more classes). Fine-tuning typically requires a lower learning rate than training from scratch. Set a lower learning rate to adapt the model to the dataset. Some earlier layers may be frozen in the network (especially in the backbone), and only later layers (the neck and head) are trained, which are more specific to the detection task. As training progresses, more layers are unfrozen for full fine-tuning. Fine-tuning improves model accuracy for tasks involving unique object classes, where pre-trained models may not perform as well.

Objective function

An input label video vector $V = [v_0, v_1, \ldots, v_c], v_i \in \{0,1\}$is given. The confidence score of all emotional classes "c" is given by the proposed model as $G = [g_0, g_1, \ldots, g_c], g_i \in [0,1]$. The probability that the model assigns to indicate how confidently it assigns the input to a particular class is called a confidence score. The binary cross-entropy loss function is an end-to-end learning strategy when training the network to adapt to the multi-label classification model. Weights are applied to improve the loss of positive samples due to class imbalance issues in the datasets. The loss of each training sample is computed as follows:

$$Loss\,(v, g) = \sum -\frac{1}{c}[w_i.v_i.\log(g_i) + (1 - v_i).\log(1 - g_i)] \tag{19}$$

$$where\ w_i = \frac{neg_i}{pos_i} \tag{20}$$

For class i, the numbers of negative and positive samples are denoted by neg_i and pos_i.

Optimization for better results

To reduce the computational cost of the M-*fus*HER model, the model is optimized with frame selection and compression techniques. Using frame selection and compression together creates a highly efficient pipeline for video-based emotion recognition in the M-*fus*HER. They work synergistically to reduce redundant data, lower memory usage, and speed up processing while maintaining performance.

Frame Selection Optimization is intelligently choosing the most relevant frames from a video to maximize information gain while minimizing computational cost. Instead of processing every frame, optimization techniques ensure that only keyframes—those carrying the most meaningful visual information—are selected. This is crucial in M-*fus*HER, which relies on CNN-mLSTM, ALBERT, and YOLOv6 for emotion recognition, as it helps reduce redundant processing, memory usage, and inference time while maintaining accuracy. Frame selection optimization is essential for balancing accuracy and efficiency in video-based emotion recognition. By combining motion analysis, deep learning, and scene detection, M-*fus*Her can process only the most meaningful frames, reducing computational costs while maintaining recognition performance. By optimizing frame selection, the number of frames processed is reduced while maintaining the performance. Assuming a video has F frames, YOLOv6 has a computational complexity of O(Y) per frame, and then the total cost is given as:

$$C_{YOLO} = F.O\,(Y) \tag{21}$$

After frame selection optimization, let F' be the number of selected frames where F'<< F, then the new cost is:

$$C'_{YOLO} = F' .O(Y) \tag{22}$$

The reduction factor becomes less than 1 as:

$$\frac{C'_{YOLO}}{C_{YOLO}} = \frac{F'}{F} \ll 1 \tag{23}$$

Hence, frame selection helps reduce the overall model's computational cost.

Implementation and results

Datasets used

The experiments were carried out on two datasets: Interactive Emotional Dyadic Motion Capture (IEMOCAP) [32, 33], Multimodal Opinion Sentiment and emotion Intensity (MOSEI) [34, 35], and Multimodal EmotionLines Dataset (MELD) [36, 37] datasets. Table 2 presents the characteristics of all the datasets used. IEMOCAP consists of 152 dialogue-recorded videos in which two speakers converse with several utterances in each video. The dataset was initially tagged with nine different emotions. Only six categories are retained due to the imbalance: angry, pleased, enthusiastic, sad, irritated, and neutral. The dialogue videos are divided into 7380 sub-clips. This dataset is split into 75%–10%–15% training-validation-testing sets. MOSEI is one of the most enormous datasets for multimodal emotion recognition. This dataset consists of 20,477 total video samples. It has six emotions: happy, sad, anger, fear, surprise, and disgust. This dataset is also split in the same manner as IEMOCAP is split. The Multimodal EmotionLines Dataset (MELD) is an extended and enhanced version of the EmotionLines dataset. It contains the same dialogue instances as EmotionLines but incorporates audio and visual modalities alongside text. MELD includes over 1,400 dialogues and 13,000 utterances from the TV series *Friends*, featuring multiple speakers. Each utterance in a dialogue is labeled with one of seven emotions: Anger, Disgust, Sadness, Joy, Neutral, Surprise, or Fear.

Since M-fusHER is a multimodal fusion model for emotion recognition (combining text, audio, and video), it handles challenges like noise, missing modalities, and incomplete features. Automatic Speech Recognition (ASR) errors introduce typos or missing

Table 2 The characteristics of IEMOCAP and MOSEI datasets with three modalities visual(v), text(t), and audio(a)

Dataset	IEMOCAP	MOSEI	MELD
Modalities	v, t, a	v, t, a	v, t, a
Training	5262	14,556	10,325
Validation	740	1776	1400
Testing	1378	4145	1275
Total	7380	20,477	13,000
Labels	Anger, frustration, excitement, happiness, sad, neutral	Anger, fear, disgust, happy, sad, surprise	Anger, disgust, sadness, joy, neutral, surprise, or fear

words in the textual data that cause informal language, sarcasm, and slang, which can mislead emotion classification. ALBERT uses self-attention, so the model infers meaning from the surrounding context even if certain words are misrecognized. Background noise (e.g., crowd sounds, static) affects speech feature extraction, which leads to low-quality recordings and reduces model accuracy. CNN-mLSTM uses Mel-Frequency Cepstral Coefficients (MFCC) instead of raw waveforms. Low-light conditions, motion blur, and occlusions affect facial expression recognition. Grad-CAM attention maps can be used to ensure the model focuses on the most expressive facial regions. Multimodal Hierarchical Attention Transformer Encoder (MHATE) is used to combine all the modalities. It helps effectively combine text, audio, and video information while addressing issues like modality imbalance, noise, and missing data. In this study, MHATE captures essential features in each modality and helps the model learn interactions between modalities. MHATE can improve M-*fus*HER's performance by optimizing how different modalities interact, handling noisy/missing data, and enhancing contextual understanding. The hierarchical attention mechanism of MHATE assigns higher weights to emotion-relevant features in each modality. It uses self-attention to capture salient words in text, important speech segments, and facial expressions. For "I didn't mean to hurt you" (text) + sad tone (audio) + neutral face (video)- MHATE's attention mechanism focuses more on tone in speech than the neutral face. Simple fusion may not capture deep interactions between modalities (e.g., sarcasm in text + tone in audio). Cross-Modal Attention of MHATE learns dependencies between text, audio, and video instead of treating them separately. The adaptive modality weighting mechanism of MHATE dynamically adjusts how much each modality contributes to the final prediction. For "Wow, that was amazing" (text) with flat tone (audio)- MHATE lowers the weight of text (which suggests excitement) and focuses on speech tone, improving sarcasm detection. Real-world data has occluded faces, low-quality audio, or ASR errors in text. Modality Dropout Training of MHATE learns to predict emotions even if one or more modalities are missing. If it is missing, the Cross-Modal Knowledge Transfer of MHATE extracts facial cues from speech instead of video. For example, if the video feed is unavailable, MHATE extracts emotional clues from the tone in audio and emotional keywords in text. It Adjusts modality weights, reducing reliance on missing video. MHATE uses transformer-based memory networks to track emotional context across multiple utterances. It helps capture emotional shifts and speaker consistency in dialogues. For "I love this… just kidding", MHATE recognizes emotion reversal by tracking the previous sentence.

Metrics used for evaluation

The performance evaluation metrics in this experiment are F1-score and accuracy. The standard accuracy metric is replaced by average weighted accuracy (AvgWA) due to the imbalance in datasets [38]. Accuracy is a combination of sensitivity and specificity. It is the ratio of predicted test findings to total tests conducted. However, one significant drawback of accuracy is its inability to differentiate between errors when specificity and sensitivity have the same values. For example, suppose specificity has a value of 0.8 and sensitivity has a value of 0.2. In that case, accuracy will be 0.5, and if the reverse happens, specificity has a value of 0.2, and sensitivity has a value of 0.8, then the accuracy will be 0.5. To solve these limitations, Liu Ying et al. [38] introduced the concept of average weighted accuracy, which is used in this study for better performance evaluation of the

proposed model. In datasets where some classes are more frequent than others, a simple average accuracy might overestimate the model's performance by focusing more on the dominant classes. Average weighted accuracy solves this by assigning more weight to the performance of underrepresented classes. It provides a better understanding of the model's performance across all classes, especially when classes are unequal.

In evaluating the performance of the proposed M-fusHER model, some common evaluation measures that have been widely applied in multimodal emotion recognition studies include accuracy (Acc), precision (Prec), recall (Rec), and F1-score have been employed. These measures give an equal consideration to classification performance because both the correct prediction count and the specific error. Besides these traditional metrics, we also used average weighted accuracy (AvgWA) and weighted accuracy (WAcc) that are especially useful when testing models on an imbalanced dataset, like IEMOCAP, MOSEI and MELD. The equations of AvgWA and WAcc have been based on Liu et al. [39] and other similar preceding studies on multimodal assessment.

Average weighted accuracy is the average of weighted accuracy. The definition of weighted accuracy (WAcc) is the accuracy that has been modified to account for the relative importance (r_{el}) of false-positive versus false-negative mistakes [39].

$$WAcc = \frac{[r_{el}p\,(specificity) + (1-p)\,sensitivity]}{r_{el}p + 1 - p} \tag{24}$$

where p is the prevalence of the classifying object.

The AvgWA is calculated using the relevant prevalence range. The relevant prevalence range is given as $p \in [c, d]$, where c and d are the lower and upper bounds for the prevalence range, respectively.

$$AvgWA = a_1 Sensitivity + a_2 Specificity \tag{25}$$

$$a_1 = \frac{1}{1-r_{el}} - \frac{r_{el}}{(d-c)(1-r_{el})^2} ln\left(\frac{r_{el} + d(1-r_{el}}{r_{el} + c(1-r_{el}}\right) \tag{26}$$

$$a_2 = \frac{r_{el}}{(1-r_{el})(d-c)} ln\left(\frac{r_{el} + d(1-r_{el})}{r_{el} + c(1-r_{el})}\right) - \frac{r_{el}}{1-r_{el}} + \frac{r_{el}^2}{(1-r_{el})^2(d-c)} ln\left(\frac{r_{el} + d(1-r_{el})}{r_{el} + c(1-r_{el})}\right) \tag{27}$$

Where a_1 and a_2 are the constants that can be calculated with the help of Eqs. 23 and 24 by using c, d, and r_{el}. The AvgWA has two advantages over the traditional accuracy metrics. First, consideration is given to the relative merits of specificity and sensitivity. Secondly, it makes an evaluation more flexible in cases where a single number cannot adequately summarise the prevalence.

Instead of evaluating a model's overall performance based on accuracy, the F1 score is an alternative machine learning assessment metric that elaborates on the model's performance inside a class. The F1 score aggregates accuracy and recall using their harmonic mean; hence, optimizing the F1 score entails simultaneously optimizing memory and precision. The F1 score is computed as follows:

$$Pre = \frac{True\ Pos}{True\ Pos + False\ Pos} \tag{28}$$

$$Rec = \frac{True\ Pos}{True\ Pos + False\ Neg} \tag{29}$$

$$F1 = \frac{2 * Pre * Rec}{Pre + Rec} \tag{30}$$

Implementation

ALBERT implementation

The profounded model is implemented with the help of Pytorch environment version 1.9.1 [40]. A pre-trained ALBERT model has been implemented [41] to extract features from the text translations. ALBERT (A Lite BERT) is a lightweight version of BERT (Bidirectional Encoder Representations from Transformers) designed to be more efficient while maintaining strong performance on NLP tasks. ALBERT reduces the model size and computational requirements compared to BERT, making it suitable for scenarios where resources are constrained or when speed is crucial. Using ALBERT to extract features from text translations involves leveraging its ability to generate contextualized embeddings for text. It can be helpful in various downstream tasks like translation evaluation, cross-lingual tasks, or integration with other models. The ALBERT model has been finetuned to improve its feature extraction performance.

ALBERT reduces the number of parameters by sharing weights across all transformer layers. Instead of having unique weights for each layer, ALBERT reuses the same set of weights, significantly reducing the model size. ALBERT decouples the embedding size from the hidden layer using a smaller embedding matrix, improving parameter efficiency. ALBERT uses SOP to understand inter-sentence relationships better. SOP identifies whether two consecutive sentences are in the correct order or swapped. The architectural optimizations in ALBERT significantly reduce memory requirements and speed up training and inference compared to BERT. ALBERT helps convert extracted facial expressions to textual descriptions and then classifies text descriptions into emotion categories.

The maximum length of the text has been confined to 100 words per second. Every 1 s, a sample of the video frames is taken for each experiment, and the pre-trained CNN-mLSTM model is used to recognize and extract facial regions. The resulting cropped faces are scaled to 64 × 64 and used as the visual backbone's input. Simultaneously, audio signals transform into Mel-spectrogram format and are segmented into arrays of chunks utilizing a time window of 400ms for the audio modality. The features of audio and image frames from videos are extracted with the help of CNN-mLSTM. They are passed as input to the finetuned YOLOv6 object recognition model, where the model classifies the emotions. For each dataset, several encoders and decoders that need to be employed by the audio encoder, image encoder, fusion module, and embedding network are provided in the ablation study. Each transformer encoder and decoder have four heads (m = 4), an FF network dimension of 2048, and a hidden size of 256 (s = 256).

The complete model is trained using the Adam optimizer [42]. The RTX8000 GPU with 16GB RAM is used to train the model, and the batch size is set to 32. The model's initial learning rate is 5×10^{-5}. The model is trained for 100 epochs for both MOSEI and IEMOCAP datasets.

Implementing ALBERT with CNN-mLSTM

To merge ALBERT with CNN and Multiplicative LSTM (mLSTM) for emotion recognition from images and videos, a multi-modal architecture that combines CNN, ALBERT,

and m-LSTM along with a fusion layer is developed. CNN helps to extract the spatiotemporal features from the extracted facial expression features from the images. ALBERT helps in processing the textual descriptions from the features extracted. mLSTM captures sequential emotion variations from the extracted features. the fusion layer merges all three before the final classification. In this study, the fusion layer helps merge CNN and ALBERT outputs before the final classification.

Assuming X_t is the text input and X_v is the image/video input. The features from both modalities are extracted separately. ALBERT processes textual data and extracts the hidden state of the [CLS] token as:

$$H_t = f_{ALBERT}\left(X_t\right) \in \mathbb{R}^{d_t} \tag{31}$$

where H_t is a d_t-dimensional feature vector from ALBERT.

CNN extracts spatiotemporal features from the image/video frame as:

$$H_v = f_{CNN}\left(X_v\right) \in \mathbb{R}^{d_v} \tag{32}$$

where H_v is a d_v-dimensional feature vector from CNN.

For video sequences, CNN features are passed through mLSTM to capture temporal dependencies as:

$$H'_v = f_{mLSTM}\left(H_v\right) \in \mathbb{R}^{d_v} \tag{33}$$

where H'_v represents the temporally processed video feature.

Merging ALBERT and CNN-mLSTM with YOLOv6

Fine-tuned YOLOv6 is used to detect faces and extract emotion-related facial features. Instead of just using raw images, YOLOv6 is used to focus on relevant facial regions as:

$$Detection = f_{YOLOv6}\left(X_v\right) \tag{34}$$

where X_v is inputted into the video frame.

The cropped face is extracted and passed to CNN as:

$$H_v = f_{CNN}\left(cropped\ face\right) \tag{35}$$

where H_v is the feature vector of size d_v.

Temporal features are processed using mLSTM as:

$$H'_v = f_{mLSTM}\left(H_v\right) \in \mathbb{R}^{d_v} \tag{36}$$

where H'_v represents the spatiotemporal features of facial expressions.

Now, merge the extracted text features H_t and image/video features H'_v into a single representation. This can be done with three methods: concatenation-based fusion (known as early fusion), multiplicative fusion (known as Hadamard product), and attention-based fusion (known as adaptive weighting). In this study, the concatenation-based fusion method is used. This is a simple method to concatenate both feature vectors as:

$$H_{fusion} = \left[H_t; H'_v\right] \tag{37}$$

where $H_{fusion} \in \mathbb{R}^{d_v+d_t}$, and is a combined feature vector. This is passed through a fully connected layer for classification as:

$$Y = \sigma\left(WH_{fusion} + b\right) \tag{38}$$

where W is the weight matrix, b is the bias, and σ is the activation function (softmax in this study).

After fusion, H_{fusion} is passed through a softmax classifier as:

$$Y = softmax(W_c H_{fusion} + b_c) \tag{39}$$

where Y is the predicted emotion class, W_c and b_c are trainable parameters.

The whole process can be represented as:

$$Y = softmax(W_c\left(f_{fusion}\left(H_t, H_v'\right)\right) + b_c) \tag{40}$$

where f_{fusion} is the chosen fusion strategy (concatenation).

Results

The outcomes of the proposed model for the IEMOCAP dataset are displayed in Table 3. When comparing the findings at each particular modality, the profounded model performed better in terms of AvgWA and F1 score. Tables 4 and 5 display the outcomes of the suggested model on the MOSEI and MELD datasets. Table 6 shows that the proposed model performs better in accuracy than any other SOTA model. The suggested model, M-*fus*HER, considerably improved performance on emotion recognition. These results validate the proposed model's effectiveness in integrating multimodal video input and obtaining robust representations for identifying multi-label emotions. The findings also corroborate the idea that multimodal learning is more effective than unimodal learning and that training models using unprocessed data rather than manually crafted features is preferable. Highlighted values in Table 3 to Table 6 indicate the highest accuracy or F1-score achieved for a given emotion across all evaluated models, signifying the superior performance of that modality or the proposed M-fusHER model.

The model is compared with different existing methods. The profound model is compared with Iram Hina et al. [43]model, the accuracy achieved was 61.68%, with Juan-pablo et al. [44] where the accuracy was77.6%, and Ayushi et al. [45]. The comparison has been made based on the usage of the same datasets. Table 6 presents the comparison. Table 7 presents the inference time analysis of M-*fus*HER under different configurations, and Fig. 3 presents the runtime vs. accuracy trade-offs of the proposed model.

The model is also evaluated in a real-time environment. A random video is used to evaluate the suggested model, M-*fus*HER. A video has a person's face with different emotions embedded in it. The suggested model can categorize various emotions. The assessment of various emotions in this video by M-*fus*HER is displayed in Fig. 4. The model classifies more than one emotion in a present frame, like in one of the images in Fig. 4, the model can classify 'sad' and 'angry' emotions both at the same time. This shows the proposed model is much more reliable and advanced than the stated SOTA existing models. The confusion matrix for IEMOCAP, MOSEI, and MELD datasets is depicted in Fig. 5. Figure 6 illustrates the accuracy of the epoch curve and loss to epoch curve for the M-*fus*HER model. The curves are evaluated for 25 epochs only, and for 100 epochs, the time consumed by the model was more. This is one of the limitations of this model. The model is time-consuming. This limitation will be worked upon in the future. Therefore, for 25 epochs, the model's accuracy and loss are evaluated.

Table 3 The results on the IEMOCAP dataset with the proposed model are compared with the results obtained at each modality

Model	Anger		Frustrated		Excited		Happiness		Sad		Neutral	
	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1
Audio-CNN-mLSTM, YOLOv6	91.2	78.9	90.2	87.2	93.4	82.3	92.3	84.5	95.4	90.2	89.7	87.7
Image-CNN-mLSTM, YOLOv6	89.6	88.9	95.5	90.1	92.4	85.6	93.5	82.3	90.1	87.5	91.5	79.8
Text- ALBERT	86.4	76.5	90.4	75.4	92.3	88.1	92.3	76.5	89.1	67.8	93.4	87.6
M-*fus*HER	**92.5**	**89.7**	**95.56**	**91.2**	**94.5**	**89.6**	**93.6**	**87.8**	**95.5**	**90.8**	**94.2**	**88.5**

A comprehensive runtime performance analysis was undertaken to support the real-time performance of the proposed M-fusHER model. Tests are run on NVIDIA RTX8000 with 16 GB of RAM with an inference batch size of 32. The results indicate that the system would require one frame in approximately 16 ms, which, consequently, can be converted to an average of 28 frame/s per second (FPS). Because real-time video applications usually need at least 24 FPS, these findings prove that M-fusHER reaches real-time performance in settings with a GPU enabled. The average end-to-end time of processing was 12 s to extract features, multimodal fusion, and classification time on 10 s of (approximately 300) video clips. These results prove that the system can handle real-time video streams. It should be noted, though, that now, runtime efficiency requires the use of high-performance GPU hardware. Further optimizations (e.g., model pruning, quantization, or knowledge distillation) will be explored to reduce the number of computations needed, at the cost of accuracy, to run on resource-constrained devices such as mobile or embedded systems.

Ablation study

The ablation experiment assessed how the emotion-level embedding module and TFM influence the model's performance. Table 8 shows the contributions of all the modules. The embedding module in the first test case is removed to examine the effects of the fusion module. The IEMOCAP dataset's fusion module improves the AvgWA and F1 scores by +0.4% and +1.40%, respectively. On the MOSEI dataset, the fusion module also increased the AvgWA and F1 scores by +0.88% and +0.91%, respectively. The embedding module is linked to the fusion module to train the model for the second test scenario. This contributes to further enhancing the model's performance. The IEMOCAP and MOSEI datasets' AvgWA and F1 scores increase once the embedding level is added to the second test case. Therefore, the suggested model attained the maximum accuracy and F1 scores for both datasets when both the transformer-based module and the embedding level fusion modules were integrated into the M-*fus*HER model. The ablation study also specifies the number of transformer encoder and decoder layers ideal for feature extraction and embedding level fusion. By applying three layers of transformer encoders and decoders specifically for IEMOCAP and employing a dual-layer approach for each module in the MOSEI datasets, the suggested model produced the best results regarding accuracy and F1 score. These layers are variable with an increase in dataset size.

Discussion

The results of the proposed M-fusHER model in multimodal emotion recognition are extremely good and close to three benchmark databases (IEMOCAP, MOSEI, and MELD) and provide better results than the state-of-the-art baselines. Successful capture of complementary text, audio, and video modality features was achieved by CNN-mLSTM-based feature extraction, Transformer-based fusion mechanism (MHATE), and fine-tuned YOLOv6-based emotion classification. The ablation study also confirmed the significance of the fusion and embedding modules towards enhancing the accuracy and the robustness of recognition. M-fusHER uses simultaneous multimodal fusion both at the feature and embedding levels, leading to improved detection rates (95.45% on IEMOCAP, 88.76% on MOSEI, and 95.41% on MELD) compared to previous unimodal

Table 4 The results on the MOSEI dataset with the proposed model are compared with the results obtained at each modality

Model	Anger		Fear		Disgust		Happy		Sad		Surprise	
	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1
Audio-CNN-mLSTM, YOLOv6	90.2	79.9	91.2	86.2	92.4	84.3	93.3	85.5	94.4	89.2	90.7	88.7
Image-CNN-mLSTM, YOLOv6	88.6	84.9	94.5	89.1	93.4	87.6	92.5	84.3	91.1	86.5	92.5	78.8
Text- ALBERT	87.4	78.5	91.4	77.4	94.3	89.1	93.3	77.5	90.1	76.8	94.4	88.6
M-*fus*HER	**93.5**	**88.7**	**94.56**	**91.2**	**95.5**	**88.6**	**94.6**	**87.8**	**95.45**	**89.8**	**95.2**	**85.5**

and bimodal methods. More recent methods, such as multimodal transformers or memory-based fusion, have shown considerable promise but are unable to handle dirty or missing modalities. Compared to the hierarchical attention and modality dropout mechanisms, the M-fusHER can maintain a stabilized contact in conditions that impair it with a single modality. This advantage makes it particularly useful within a real-life scenario, where data quality is not necessarily identical.

An essential advantage of M-fusHER is its capability to work in a real-time video setting, as evidenced by the analysis of the live video sequences. Contrary to most existing research, which uses only offline data sets, M-fusHER managed to detect multiple emotions in overlapping frames (e.g., sadness and anger), which indicates that the model can deal with the intricacies of feelings as they are expressed in the real world. However, its disadvantage is that it is potentially computationally intensive, especially when longer video sequences are needed. Proposed frame selection and compression strategies minimise this cost; however, a future research direction is to optimise the computation efficiency. In addition to the competitive performance of the proposed M-fusHER framework on benchmark data, there are numerous significant real-world applications of this framework in a wide range of domains:

a. Intelligent Tutoring Systems (ITS): Casting Emotion-conscious ITS can react to affective learners' states of confusion, frustration, or excitement. With the implementation of M-fusHER in the tutoring system, the instructing process can become more individual, and precise content is recommended, informing an educator that he or she should intervene, which enhances the learning experience. Previous research in affective computing has shown that multimodal emotion recognition has a role in education to support adaptive and personalized learning [1].
b. Customer Behaviour Analysis: Real-time emotion recognition when interacting with chatbots, virtual assistants, or call centers can be important in the retail and service sectors. The audio-visual-textual signals reflecting dissatisfaction, hesitation, or satisfaction can be identified by M-fusHER and used to engage customers proactively and empathically. This type of affect-based analytics has already been discussed in the context of multimodal sentiment analysis and customer interaction research [9].
c. Digital Mental Health Monitoring: M-fusHER can be embedded into telemedicine software and wellness applications to track low-level multimodal emotional signals. This could help clinicians to identify the presence of stress, anxiety, or depression in the initial stages of virtual consultation and provide a means of ongoing mental health assessment beyond the clinical environment. Recent developments of multimodal models of health and well-being monitoring prove the potential influence of emotion recognition in both clinical and non-clinical settings [13].
d. Affective User Interface: By providing emotion recognition technology to smart devices, VR/AR platforms, or assistive technologies, human-computer interaction can become more natural and responsive. M-fusHER allows dynamically changing user interfaces based on the emotional state of the user and makes them more usable and satisfactory. Past attempts at affective computing have confirmed that multimodal modelling can be used to build emotion-sensitive systems [18].
e. Security and Surveillance: Real-time stress, fear, or aggression detection in surveillance video streams can be used to help improve the safety of the public. M-fusHER is a

Table 5 The results on the MELD dataset with the proposed model are compared with the results obtained at each modality

Model	Anger		Fear		Disgust		Happy		Sad		Surprise	
	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1	AvgWA	F1
Audio-CNN-mLSTM, YOLOv6	91.2	80.9	92.2	87.2	93.4	85.3	94.3	86.5	95.4	90.2	91.7	89.7
Image-CNN-mLSTM, YOLOv6	89.6	85.9	95.5	90.1	94.4	88.6	93.5	85.3	92.1	87.5	93.5	79.8
Text- ALBERT	88.4	79.5	92.4	78.4	95.3	88.1	92.3	78.5	91.1	77.8	95.4	87.6
M-*fus*HER	**94.5**	**89.7**	**95.56**	**90.2**	**93.5**	**89.6**	**93.6**	**88.8**	**94.45**	**88.8**	**94.2**	**84.5**

multimodal fusion that helps to identify potentially dangerous situations early in sensitive environments such as airports, schools, and big events. Recent studies have also reported similar uses of multimodal emotion recognition in safety-critical settings [21].

Although M-fusHER was found to be very accurate in various data sets, it requires heavy architectures (CNN-mLSTM and Transformers) and prevents faster training and inference times. This might be a hindrance to resource-constrained devices. The second research wave, lightweight and knowledge distillation with model pruning, will be linked to a no-loss performance optimization. In addition, the assessment will be additionally grounded on even more diverse and noisy real-world data (e.g., in-the-wild video data, cross-cultural data of emotions corpora). Textual cues can significantly enhance the perception of multimodal emotions, but their application in the natural world may be limited or unreliable where automatic speech recognition (ASR) can make errors, or where textual subtitles are absent or the conversational content is noisy. This limits the practical value of models, which can be highly dependent upon clean text inputs. To address this issue, the proposed M-fusHER framework utilizes the Multimodal Hierarchical Attention Transformer Encoder (MHATE), which can dynamically change how much each modality contributes to the processing process depending on the availability and quality of the data. The construction allows the model to be resilient when text information is partially missing or destroyed, and instead relies more on audio-visual techniques. Nevertheless, the evaluation of M-fusHER without or with noisy textual input is also a line of research worth future investigation to better represent the deployment experience.

Conclusion

In this study, a multi-label multimodal emotion recognition model, M-fusHER, is proposed. The profound model is an end-to-end combination of a transformer-based encoder and decoder module with an embedding-level fusion module. In this study, the multimodal aspects of the video data at several layers are fused simultaneously by taking advantage of the transformers' multi-head attention. Additionally, emotion-level representations were learned from fused multimodal information to enhance model performance in the multi-label identification task. The model was trained and evaluated using IEMOCAP, MOSEI, and MELD datasets. The experimental findings indicate that the suggested model performs better than other robust baselines and earlier techniques. The model is also evaluated in a real-time environment with a random video, and the results demonstrate that the developed model can classify all the emotions depicted by the person in the video; the profound model can even classify multiple emotions at a time. The supplied ablation research shows how the emotion-level embedding and fusion modules improve the model's performance. The video modality had the most expensive computational cost during experimentation due to redundant and identical frames. In the future, the model will be enhanced to remove those redundant and identical frames from the video to minimize the computational cost. In its current state, it is a time-consuming model, which will be worked on in the future. Optimizing M-fusHER for faster training and reduced computational cost requires improvements in model architecture, data efficiency, and hardware utilization. Processing every frame from video data increases

Table 6 The comparison results of SOTA methods with the proposed model

Citation	Dataset	Accuracy (%)	F1-score (%)
Iram Hina et al. [34]	IEMOCAP	61.68	59.86
Juanpablo Heredia et al. [20]	IEMOCAP	77.6	75.65
Aayushi Chaudhari et al. [10]	CMU-MOSEI	87.6	-
Yuanyuan et al. [9]	IEMOCAP, MELD	72	67.89
Jintao et al. [12]	IEMOCAP, MELD	64	63.45
M-*fus*HER	**IEMOCAP, MOSEI, MELD**	**95.45, 88.76, 95.41**	**94.8, 87.9, 94.5**

Table 7 The inference time analysis of M-fusHER under various configurations

Configuration	Model Depth	Parameters (M)	Input Modality	Avg.time per frame (ms)	FPS	Avg. Time per 10 s Clip (s)	Accuracy (%)	Notes
Text-Only	Shallow (8)	35	T	15	21	7	93.2	Lower latency, limited accuracy
Audio-only	Shallow (8)	34	A	12	22	8	93.7	Sensitive to background noise
Visual-only	Medium (12)	40	V	15	27	10	94.5	Computationally heavier
Text+Audio	Medium (12)	39	T+A	16	26	9	94.9	Balanced trade-off
Audio+Visual	Medium (12)	39	A+V	16	26	9	95.2	Noisy environments affect speed
Text+Visual	Medium (12)	39	T+V	16	26	9	94.91	Fusion adds moderate overhead
Full (Text + Audio + Visual)	Deep (16)	40	T+A+V	16	28	10	95.45	Best accuracy, highest overhead
Full (Pruned/ Lightweight Variant)	Deep (12)	37	T+A+V	15	28	9	95.45	Optimized for speed

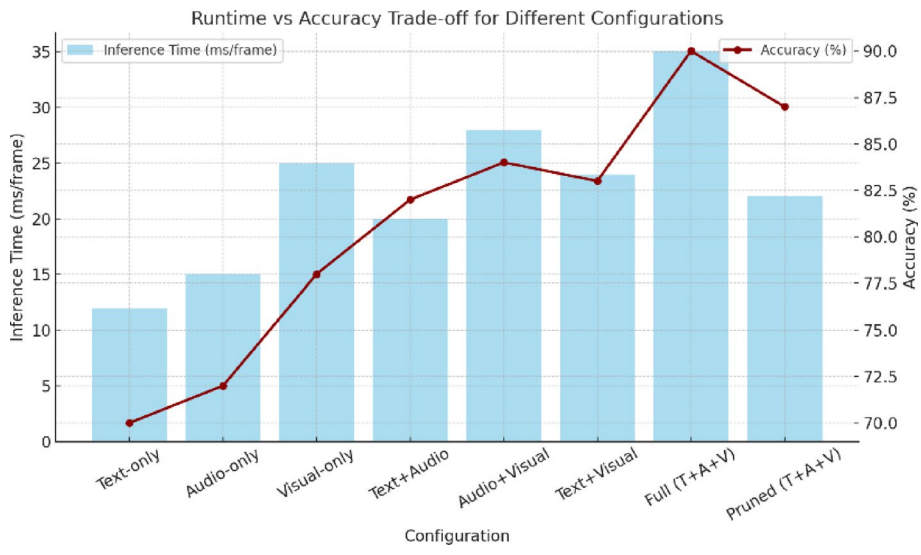

Fig. 3 Runtime vs. accuracy trade-offs of M-*fus*HER

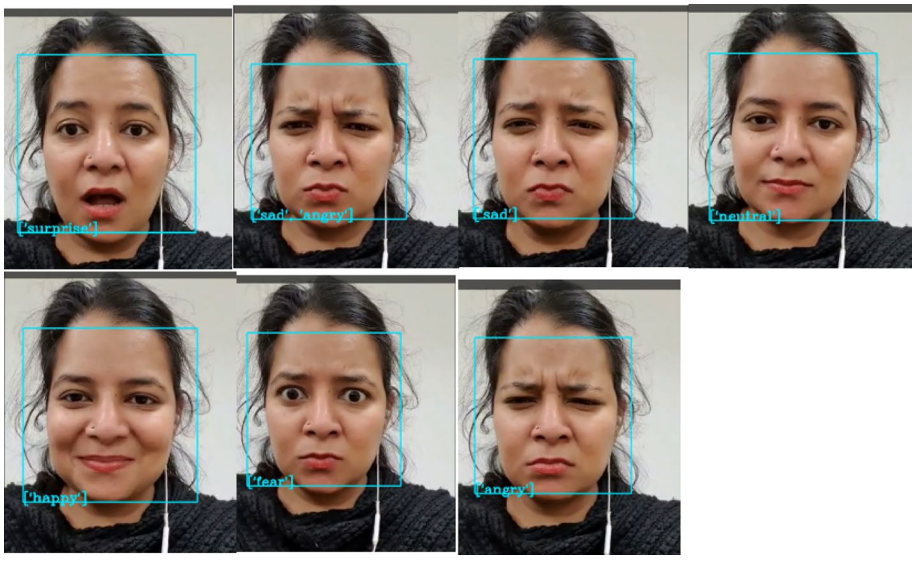

Fig. 4 Real-time evaluation of the proposed model M-*fus*HER

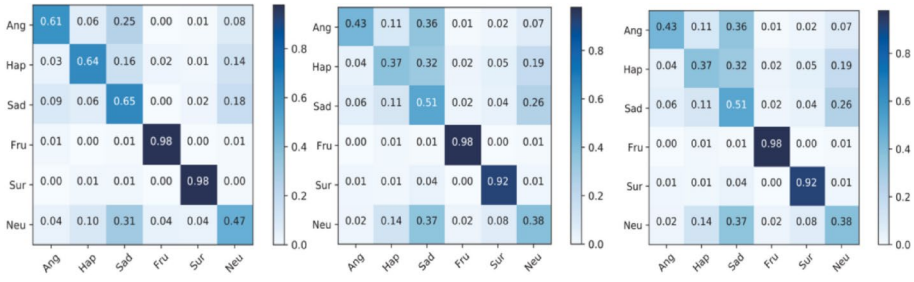

Fig. 5 Confusion matrix evaluated for IEMOCAP, MOSEI, and MELD datasets

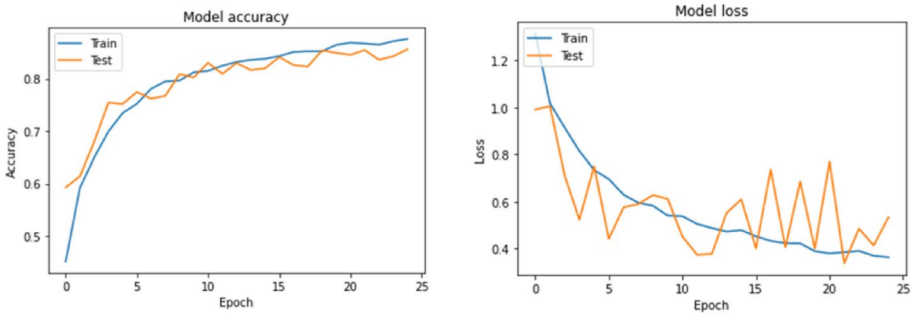

Fig. 6 Proposed model's accuracy and loss curves w.r.t 25 epochs

computational cost. Instead of heavy networks, use efficient architectures for feature extraction. Instead of concatenating all features, ignore unreliable modalities to reduce unnecessary computations. Apply modality dropout to reduce dependency on all three modalities at every step.

Table 8 Component-wise ablation experiment on the IEMOCAP and MOSEI datasets using M-*fus*HER

MODEL	IEMOCAP		MOSEI	
	AvgWA	F1	AvgWA	F1
Text- ALBERT	85.67	76.87	87.78	65.76
Image frames – YOLOv6	90.56	85.67	91.23	87.56
Audio – YOLOv6	93.45	90.23	89.34	85.67
Transformer-based fusion only	94.56 (+0.4%)	91.23 (+1.40%)	94.34 (+0.88%)	90.89 (+0.91%)
Transformer-based fusion + embedding fusion	95.45%	91.34	95.43	90.98

Acknowledgements
Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R817), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Author contributions
All authors contributed equally in all aspects of the conceptualization of the paper, its design, material preparation, data collection, analysis, and the first draft of the manuscript.

Funding
No funding is received for the study.

Data availability
https://sail.usc.edu/iemocap/, https://www.kaggle.com/datasets/gnurtqh/cmu-mosei, https://www.kaggle.com/datasets/zaber666/meld-dataset/code.

Declarations

Competing interests
The authors declare no competing interests.

Author details
[1]Department of Computer Science and Applications, Maharshi Dayanand University, Rohtak, India
[2]Faculty at School of Information Technology, Vivekananda Institute of Professional Studies-Technical Campus, Delhi, India
[3]Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Rajpura 140401, Punjab, India
[4]Department of Mechanical Engineering and Renewable Energy, Technical Engineering College, The Islamic University, Najaf, Iraq
[5]Department of Computer Sciences, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia
[6]Department of Mathematics, Hawassa University, P.O. Box 05, Hawassa, Ethiopia
[7]Department of Industrial and Systems Engineering, College of Engineering, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

Received: 26 April 2025 / Accepted: 8 October 2025

Published online: 18 November 2025

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